Wiring Guide — Automated Test Station

> **Print-ready PDF:** `docs/pdf/wiring-guide.pdf`

This guide covers all wiring for the Juniper Automated Test Station thermal rig: the RPi 4 HMI host, Siemens LOGO! PLC, 4× MAX31855 thermocouple amplifiers, SSR-driven heater, servo-controlled vents, and DUT interface relays.

For instrument-specific wiring (Vitrek V71 HiPot, Siglent SDL1020X-E DC Load), see the respective instrument manuals.

System Block Diagram

```
                         ┌──────────────────────────────────────────────────────┐
                         │          DIN RAIL ENCLOSURE (24V DC)                 │
                         │                                                      │
    ┌──────────────┐     │  ┌──────────────────┐   ┌──────────────────────┐     │
    │ RPi 4 HMI    │◄─LAN─► LOGO! 12/24RCE │   │  24V PSU (DIN rail) │     │
    │ (or NUC PC)  │     │  │ Modbus TCP:502   │   │  Meanwell HDR-60-24  │     │
    └──────┬───────┘     │  │                  │   └──────────┬───────────┘     │
           │ USB         │  │ Q1 ──────────────────►SSR → Heater (120VAC)        │
           │             │  │ Q2 ──────────────────►Servo power relay            │
    ┌──────▼───────┐     │  │ Q3 ──────────────────►DUT power relay              │
    │ V71 HiPot    │     │  │ Q4 ──────────────────►Alarm beacon                 │
    │ SDL1020X-E   │     │  │ I1 ◄──────────────────E-stop (NC)                  │
    └──────────────┘     │  │ I2 ◄──────────────────Door interlock (NC)          │
                         │  │ I3 ◄──────────────────HW overtemp cutout (NO)      │
           │ SPI         │  │ I4 ◄──────────────────Manual start (NO)            │
    ┌──────▼───────┐     │  │ I5 ◄──────────────────Manual stop (NC)             │
    │ 4× MAX31855  │     │  └──────────────────┘                                │
    │ TC amplifiers│     │                                                      │
    │ TC1 Ambient  │     │  ┌───────────────────────────────────────────────┐   │
    │ TC2 DUT      │     │  │ CHAMBER                                       │   │
    │ TC3 Heater   │     │  │ • 150W AC cartridge heater (SSR-switched)     │   │
    │ TC4 Exhaust  │     │  │ • Servo A — intake vent valve                 │   │
    └──────────────┘     │  │ • Servo B — exhaust vent valve                │   │
                         │  │ • Bimetallic thermostat 120°C (HW cutout)     │   │
    RPi GPIO 12 ──────────────►SSR signal input                           │   │
    RPi GPIO 13 ──────────────►Servo A PWM                                │   │
    RPi GPIO 18 ──────────────────►Servo B PWM                            │   │
                         │  └───────────────────────────────────────────────┘   │
                         └──────────────────────────────────────────────────────┘
```

Wire Color Convention

Matches the `automated-drill` project's convention. Use these colors from your wire kit:

Color	Signal type
Red	24V positive, mains live (L), SSR load output
Black	GND, mains neutral (N), common return
Orange	PWM signals (heater SSR control, servo PWM)
Green	Digital outputs from LOGO! (relay coil drive lines)
Blue	Digital inputs to LOGO! (E-stop, door, sensors)
Purple	SPI signals (MOSI, MISO, SCLK) and chip-selects
Yellow	I^2C / UART communication lines
White	Shield / chassis GND where needed for noise

Power Rails

Rail	Source	Feeds
120V AC	Mains (wall outlet)	SSR load side → Heater only. Never inside the DIN rail box.
24V DC	Meanwell HDR-60-24 DIN PSU	LOGO! PLC, relay coils, servo power rail (via Q2)
5V DC	RPi USB-C power supply	RPi 4 board, MAX31855 modules (via 3.3V LDO onboard)
3.3V DC	RPi GPIO 3.3V pin (pin 1/17)	MAX31855 VCC, servo signal pull-ups
GND	Single star point at DIN rail	All grounds — RPi, LOGO!, PSU, SSR signal

> **Safety:** The heater is mains-powered. All mains wiring must be in conduit, properly terminated, and fused. The SSR provides electrical isolation between the low-voltage control circuit and mains. Never expose mains terminals.

RPi 4 → MAX31855 Thermocouple Amplifiers

Uses SPI0 hardware bus. Four MAX31855 modules share SCLK and MISO; each has its own CS line.

RPi Pin	BCM GPIO	Signal	Wire	MAX31855 Pin	Notes
23	GPIO11	SCLK	Purple	CLK (all 4 modules)	SPI clock — daisy-chain to all CSs
21	GPIO9	MISO	Purple	DO (all 4 modules)	Data out — daisy-chain to all DOs
24	GPIO8	CE0	Purple	CS — TC1 (Ambient)	SPI chip-select 0
26	GPIO7	CE1	Purple	CS — TC2 (DUT)	SPI chip-select 1
22	GPIO25	GPIO out	Purple	CS — TC3 (Heater)	Software CS — set GPIO25 as output
18	GPIO24	GPIO out	Purple	CS — TC4 (Exhaust)	Software CS — set GPIO24 as output
1	3.3V	VCC	Red	VCC (all 4 modules)	3.3V — MAX31855 is 3.3V native
9	GND	GND	Black	GND (all 4 modules)	Common GND

MAX31855 → Thermocouple terminals

Each MAX31855 module has two screw terminals for the K-type thermocouple wires. K-type convention: **yellow = positive (+)**, **red = negative (-)** (US ANSI standard).

TC Channel	Location	Terminal +	Terminal -
TC1	Chamber ambient air	Yellow	Red
TC2	DUT mounting surface	Yellow	Red
TC3	Heater element body	Yellow	Red
TC4	Exhaust duct / vent	Yellow	Red

> **Cold junction:** The MAX31855 has an internal cold-junction compensator. Mount all four modules in the same thermal environment (same DIN sub-rail or common block) to minimise cold-junction error between channels.

Optional: Thermocouple transmitter for direct LOGO! PLC input

For the prototype, all temperature readings go through the MAX31855 → RPi path. This is sufficient for software-controlled PID and monitoring.

If you want the LOGO! PLC to **independently** read temperatures — for example to implement a hardware temperature interlock in ladder logic without relying on the RPi — you need a **thermocouple transmitter** (also called a TC signal conditioner). This converts the K-type thermocouple's millivolt signal into a 0–10V or 4–20 mA signal that the LOGO!'s analog inputs (AI1–AI4) can read directly.

Parameter	Requirement
Input	K-type thermocouple
Output	0–10V DC or 4–20 mA (match LOGO! AI input range)
Range	0–200°C minimum
Form factor	DIN rail mount preferred (fits alongside LOGO! in enclosure)
Example	Autonics MTH-D-VK (K-type, 0–10V out) ~$25 · or generic OMEGA TX93A ~$35
Qty	1 per thermocouple channel you want the PLC to read independently

> **Prototype note:** Transmitters are NOT required for the first build. Add them later if you want hardware-enforced PLC temperature interlocks independent of RPi software. Flag them in the BOM as "production upgrade."

RPi 4 → Heater SSR (GPIO 12)

The RPi GPIO12 drives the signal input of a Fotek SSR-40DA solid-state relay, which switches the 120V AC heater circuit.

RPi Pin	BCM	Signal	Wire	SSR Terminal	Notes
32	GPIO12	PWM (Heater)	Orange	+ (signal+)	GPIO12 = hardware PWM channel 0
34	GND	GND	Black	- (signal-)	SSR control GND

SSR load side (mains — by a qualified electrician):

The complete mains circuit runs through four independent hardware protection layers in series. Each layer operates independently of software and independently of every other layer:

```
Mains Live (L)
  │
  ├──▸ [3A inline fuse]                    ← protects the branch circuit
  │
  ├──▸ [E-stop mushroom button, NC]         ← hardwired, bypasses all electronics
  │
  ├──▸ [LOGO! Q1 relay, NO contacts]       ← safety-gated by ladder logic
  │
  ├──▸ [Fotek SSR-40DA]                    ← controlled by RPi GPIO12 PWM
  │
  ├──▸ [Thermal cutoff fuse, 150°C, 10A]  ← clipped directly to heater body
  │     (one-time, passive, no software)
  │
  └──▸ Heater element (150W, 120VAC)
         │
        Neutral (N)
```

SSR Terminal	Connects to	Notes
1 (AC in)	Mains Live (L)	After 3A fuse, after E-stop, after Q1 relay contacts
2 (AC out)	Thermal fuse → Heater +	Thermal fuse in series between SSR output and heater terminal
Heater return	Mains Neutral (N)	Returns neutral directly — SSR switches the live side only

Thermal cutoff fuse spec:

Parameter	Value
Type	Axial thermal cutoff (TCO) fuse — one-time, non-resettable
Trip temp	150°C (gives ~30°C headroom above the 120°C bimetallic thermostat)
Current	10A rated (heater draws 1.25A — heavily derated for long life)
Mounting	Clip or zip-tie directly to the heater cartridge body — must read element temp, NOT air temp
Example part	Bourns SF-0603HIA150C-2 or Microtemp G4A15X (generic 150°C 10A axial)
Cost	~$1–2 each — keep 2–3 spares on hand

> **Important:** A blown thermal fuse means something in the safety chain failed before it. Do not simply replace the fuse and restart — investigate why the element exceeded 150°C before running again.

SSR spec: Fotek SSR-40DA (3–32V DC control, 24–480V AC load, 40A, zero-crossing). Heater: 150W 120VAC cartridge heater (e.g., Omega CIR-1012/120V or equivalent). Rated SSR load current at 150W/120V: 1.25A — well within the 40A rating.

RPi 4 → Servo Motors (GPIO 13 / GPIO 18)

Standard 50 Hz PWM servo signal. Pulse width 500–2500 µs = 0°–180°.

RPi Pin	BCM	Signal	Wire	Servo Wire	Notes
33	GPIO13	PWM (Servo A)	Orange	Signal	Vent A — intake valve
12	GPIO18	PWM (Servo B)	Orange	Signal	Vent B — exhaust valve
—	5V	VCC	Red	Power (+)	Servo power from LOGO! Q2 relay
34	GND	GND	Black	GND (-)	Common GND

> **Servo power:** Servos draw up to 500 mA at stall — do not power from RPi 5V pin (max 50 mA shared). Instead, use LOGO! Q2 relay to switch 24V → 5V step-down (or use a dedicated 5V servo BEC). LOGO! Q2 acts as a safety interlock: servos are only powered when the PLC enables Q2.

LOGO! PLC Wiring

LOGO! 12/24RCE Terminal Layout

```
   LOGO! 12/24RCE
 ┌────────────────────────┐
 │ I1  I2  I3  I4  I5   │  ← Digital Inputs (24V DC source)
 │ I6  I7  I8             │
 │                         │
 │ Q1  Q2  Q3  Q4         │  ← Relay Outputs (250V AC / 5A)
 │ Q5  Q6  Q7  Q8         │
 │                         │
 │ +24V  0V  AI1  AQ1   │  ← Power + Analog (not used here)
 └────────────────────────┘
```

Power supply connections

LOGO! Terminal	Connects to	Wire	Notes
L+ (+24V)	PSU +24V output	Red	LOGO! supply in
M (0V)	PSU 0V / GND	Black	LOGO! supply return

Digital Inputs — I1 through I5

LOGO! inputs are 24V DC source-type (0V = OFF, 24V = ON in default config). All switches are wired in series with 24V and return to the LOGO! input terminal.

LOGO! Terminal	Device	Wire	Switch type	Active state	Notes
I1	E-stop mushroom button	Blue	NC	LOW = TRIPPED	Break: E-stop pressed = I1 sees 0V = TRIPPED
I2	Door reed switch	Blue	NC	LOW = OPEN	Break: door open = I2 sees 0V = interlock
I3	Bimetallic thermostat (120°C)	Blue	NO	HIGH = TRIPPED	Make: >120°C = I3 sees 24V = HW cutout
I4	Green momentary (start)	Blue	NO	HIGH = pressed	Make: button pressed = I4 sees 24V
I5	Red momentary (stop)	Blue	NC	LOW = pressed	Break: button pressed = I5 sees 0V = stop

Wiring each input (24V DC NPN-style):

```
PSU +24V ──► [switch] ──► LOGO! In (I1–I5)
                           LOGO! 0V ──► PSU 0V (common)
```

Digital Outputs — Q1 through Q4

LOGO! relay outputs are volt-free contacts (SPDT, 250V AC / 5A max).

LOGO! Terminal	Device	Wire	Load	Notes
Q1	SSR heater relay enable	Green	SSR signal+ (3–32V DC)	24V from PSU switched via Q1 → SSR input. Q1 coil energised from LOGO! relay drive.
Q2	Servo 5V power rail relay	Green	24V→5V buck input	Enables servo power rail via a 24V→5V BEC/DCDC
Q3	DUT power relay	Green	DUT supply input	Connects/disconnects power to the DUT under test
Q4	Alarm beacon	Green	24V beacon/buzzer	Energised on fault condition from ladder logic

Q relay output wiring (for 24V DC loads):

```
PSU +24V ──► LOGO! Q (Com terminal)
LOGO! Q (NO terminal) ──► Load (SSR signal+, beacon, relay coil, etc.)
Load return ──► PSU 0V
```

LOGO! Ladder Logic Specification

> The actual ladder program is written in **LOGO! Soft Comfort** (included on the CD with the PLC). This section specifies the function blocks to implement.

Safety interlock rung (heater enable)

Heater (Q1) should energise ONLY when ALL of: - I1 = HIGH (E-stop NOT tripped, NC = normal closed = 24V present) - I2 = HIGH (Door closed, NC = normal closed = 24V present) - I3 = LOW (HW thermostat not tripped, NO = normal open = 0V) - Modbus coil Q1 commanded ON by software

Ladder rung (LOGO! FBD / LAD notation):

```
[I1 NO]──[I2 NO]──[I3 NC]──[MB coil from Modbus]──►(Q1 Heater relay)
```

Where `[I1 NO]` = normally-open contact on I1, `[I3 NC]` = normally-closed contact on I3. This implements a hardware AND gate: heater can only run if all physical conditions are met regardless of software.

Servo power rung (Q2)

```
[I1 NO]──[I2 NO]──[MB coil Q2 from Modbus]──►(Q2 Servo power relay)
```

Servo power also requires E-stop and door interlock to be satisfied.

Alarm rung (Q4)

```
[I1 NC]──OR──[I2 NC]──OR──[I3 NO]──►(Q4 Alarm beacon)
```

Alarm fires if E-stop tripped OR door opened OR HW overtemp active.

Manual start/stop integration

Use LOGO! `SR` (set/reset) block: - Set: I4 (manual start) OR Modbus coil "run" - Reset: I5 (manual stop) OR I1 NC (E-stop) OR I2 NC (door) OR I3 NO (overtemp) - Output: feeds into rung condition for Q1, Q2, Q3

Breadboard Prototype Layout

The rig interface board uses a standard full-size (830-point) solderless breadboard for the prototype phase before PCB fabrication.

What goes on the breadboard: - 4× MAX31855 modules (one per TC channel) — SPI breakout pins - Pull-up resistors (10kΩ) on each CS line (4×) - 100nF decoupling capacitor on each

MAX31855 VCC pin (4×) - SSR current-limit resistor R_SSR (100Ω, ¼W, on GPIO12 → SSR+ line) - Servo signal protection resistors R_SVO (100Ω each, on GPIO13, GPIO18)

What does NOT go on the breadboard: - Mains (120V AC) wiring — always in conduit, by a qualified electrician - LOGO! PLC — mounts on DIN rail - PSU — mounts on DIN rail - SSR — mounts on heatsink on DIN rail or enclosure wall

Component placement

Component	Location	Notes
MAX31855 #1 (TC1)	Columns 1–6, top half	VCC=3.3V, GND, CS=GPIO8, CLK=GPIO11, DO=GPIO9
MAX31855 #2 (TC2)	Columns 8–13, top half	CS=GPIO7
MAX31855 #3 (TC3)	Columns 15–20, top half	CS=GPIO25 (software CS)
MAX31855 #4 (TC4)	Columns 22–27, top half	CS=GPIO24 (software CS)
C1–C4 (100nF dec.)	Adjacent to each module VCC	VCC → cap → GND at each module
R1–R4 (10kΩ CS pull-up)	CS pin → 3.3V at each module	Prevents CS floating at startup
R_SSR (100Ω)	Columns 1–2, bottom	GPIO12 → SSR+ line
R_SVO_A (100Ω)	Columns 4–5, bottom	GPIO13 → Servo A signal
R_SVO_B (100Ω)	Columns 7–8, bottom	GPIO18 → Servo B signal

External connections to breadboard

Tap point	Wire color	Destination
+ rail (top)	Red	RPi 3.3V (pin 1)
- rail (top)	Black	RPi GND (pin 9)
CS #1 tap	Purple	RPi GPIO8 (pin 24)
CS #2 tap	Purple	RPi GPIO7 (pin 26)
CS #3 tap	Purple	RPi GPIO25 (pin 22)
CS #4 tap	Purple	RPi GPIO24 (pin 18)
SCLK tap	Purple	RPi GPIO11 (pin 23)
MISO tap	Purple	RPi GPIO9 (pin 21)
R_SSR out	Orange	SSR terminal (signal+)
R_SVO_A out	Orange	Servo A signal wire
R_SVO_B out	Orange	Servo B signal wire

Power Domain Summary

Domain	Source	Consumers	Fusing
120V AC	Mains wall outlet	Heater only (via SSR)	3A inline fuse
24V DC	Meanwell HDR-60-24	LOGO! PLC, relay coils, servo BEC input, alarm	PSU built-in 2.5A
5V DC	BEC off Q2 relay	Servo power rail (2 servos, ~1A peak)	Q2 relay 5A contact
3.3V DC	RPi onboard LDO	MAX31855 modules (4×, ~4 mA each)	RPi onboard poly
GND	Star point at DIN	All grounds tied at single point	—

Safety Notes

1. **Mains wiring:** all 120V AC wiring must be performed by or verified by a qualified electrician. Use appropriate wire gauge (14 AWG for 15A circuit, 16 AWG minimum for 10A sub-circuit), rated connectors, and conduit.
2. **Thermal cutoff fuse:** the 150°C thermal fuse is the last-resort passive protection against heater thermal runaway. It must be physically touching or clamped to the heater element body — not floating in air. It is one-time only: a blown fuse means a safety failure occurred upstream. Investigate before replacing.
3. **E-stop:** the E-stop is a physical hardware interlock wired directly into the 120V heater circuit (in series, before the SSR) AND into the LOGO! input I1. Both layers operate independently. It is NOT a software flag.
4. **HW overtemp cutout:** the bimetallic thermostat disc (120°C trip) wired to I3 is a physical hardware failsafe independent of MAX31855 readings and software PID. Never bypass or remove it.
5. **Four-layer heater protection:** the heater has four independent hardware layers (inline fuse → E-stop → LOGO! Q1 relay → 150°C thermal fuse) plus the software PID cutout. All five must be functional before energising the heater for the first time.
6. **SSR heatsinking:** the Fotek SSR-40DA must be mounted on an aluminium heatsink (≥50 cm^2) or the DIN rail enclosure wall. Without a heatsink it will overheat at sustained duty cycles.
7. **Common GND:** all power rails (24V, 5V, 3.3V, SSR signal) must share a single common GND star point. Floating grounds cause measurement noise in the MAX31855 readings.
8. **Servo signal protection:** the 100Ω series resistors on servo PWM lines limit current if a GPIO pin is accidentally set as input — prevents the servo pulling the GPIO low.

PCB Prototype Board

Once the breadboard prototype is validated, the rig interface board will be fabricated using the in-house PCB pipeline (`C:\dev\pcb-fab-pipeline`). See `docs/rig-interface-board/` for: - `NETLIST.md` — full net-by-net wiring specification - `BOM.csv` — bill of materials with vendor part numbers - `board.json` — pcb-fab-pipeline board definition